Grounding Without Corrective Control: Truth-Tracking Profiles for Large Language Models

Brett Reynolds [*]
Humber Polytechnic & University of Toronto

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Abstract

Recent work suggests that some large language model representations have content or reference. Grounding can secure either without supplying live correction routes. Answerability is the graded relation in which discrepancies can affect what a target- and task-specific arrangement produces, accepts, or withdraws. Corrective control is the stronger arrangement-level capacity in which live, sufficiently independent routes can detect and repair fresh discrepancies. This paper analyses truth-tracking (patterned support for representational success) through an arrangement's route profile (its constraining routes and their relations).

Language models are the pressure case; text-only arrangements provide a task-relative limiting case. Text-trained models inherit patterns of testimony, coherence, and prior correction. Where target-sensitive correction survives training, these can supply derivative answerability (inherited constraint); live answerability is the relation supplied by a current route for fresh discrepancies. Fluent failures should follow when a task requires independently informative access to the facts. Self-consistency, retrieval, tools, code execution, multimodal input, and feedback should help selectively. Route-by-task interactions test the distinctions. A decomposition fixed in advance earns its keep by predicting held-out route–task combinations or improving intervention choice without conceptual refitting. Surface improvement and truth-tracking improvement can come apart.

Keywords: large language models; grounding; answerability; corrective control; projectibility; truth-tracking

AI use. The large language models Anthropic Claude Opus 4.8, Fable 5, and Opus 5; and OpenAI Codex using GPT-5.5 served as drafting and editing aids throughout the preparation of this paper. I'm responsible for all theoretical claims, arguments, errors, and interpretive choices.

[*]Contact: brett.reynolds@humber.ca

1 Introduction: from grounding to corrective control

Do the representations inside large language models (LLMs) connect to the world? The question is often put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language apart from what language is about. Others reply that such systems meet some conditions on reference, or that sensory grounding can be optional for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026).

But that opposition only starts the inquiry. The debate descends from the symbol-grounding problem (Harnad, 1990): how can symbols, or symbol-like states, get meaning independent of the interpretations supplied by users? For LLMs, recent work has pluralized the issue. Pavlick (2023) treats grounding as a family of questions about symbols and use; Coelho Mollo and Millière (2026) distinguish kinds of grounding; and Chalmers (2023) separates sensory grounding from thought.

The recent literature has answered sceptical arguments one at a time, principally on the side of content. On reference, the natural histories of a language model's words can secure it (Mandelkern & Linzen, 2024), while intentions are the wrong place to look (Pepp, 2025). On linguistic competence and meaning, the argument from communicative intention fails at both premises (Attah, 2025), and facts about linguistic practice can determine meaning even if the facts that would determine mental content don't (Grindrod, 2024). At the level of the proposed target, meaning can be grounded in a corpus rather than in the world (Havlík, 2024). Ruyant (2026) reaches the further conclusion that what such systems represent is a class of appropriate linguistic production rather than the world.

I contest none of those claims, and my argument doesn't require their authors to be mistaken. Some of those authors discuss misrepresentation, error, and repair, but the grounding accounts considered here don't by themselves supply an account of the live, task-indexed routes through which a fresh discrepancy can be detected and translated into a target-improving revision.[1] Establishing content and establishing the capacity to detect and repair fresh errors are different achievements, and the second doesn't follow from the first.

Distinguishing kinds of grounding still leaves open which differences in epistemic architecture explain patterned truth-relevant success and failure. Truth-relevant success depends on STABILIZING ROUTES[2] such as perception, testimony and records, coherence constraints, measurement, intervention, checking and revision, and feedback from practical success or failure. Some bring target information into production; others integrate, check, or correct what's already there.

An ARRANGEMENT is a language model together with whatever prompts, records, tools, sensors, checkers, or reviewers are live for a particular target and task. Its PROFILE records which routes constrain it and how those routes are coupled: which can check, calibrate, or correct which others. An output is ANSWERABLE to a target to the extent that discrepancies can affect what the arrangement produces or accepts, including through later correction or withdrawal. That influence may be inherited, delayed, or supplied by external audit. The arrangement has CORRECTIVE CONTROL only when

[1] Coelho Mollo and Millière (2026) build the possibility of misrepresentation into their account, and Attah (2025) discusses conversational repair. Their treatments stop short of the live routes from discrepancy to revision at issue here, and neither is committed to denying that such routes matter.

[2] In Grindrod's (2024) reading of Millikan, a STABILIZING FUNCTION is instead conventional and inter-agent: it sits between speaker and hearer. It explains why a linguistic device keeps being reproduced, and he puts the notion to work on language models. A device can have a robust stabilizing function in Millikan's sense while preserving only the effects of the routes at issue here. That's arguably what occurs in the text-only case: training preserves effects of earlier coordination without retaining the coordinating routes themselves.

live routes let it detect and repair a fresh discrepancy.

The arrangement bears the profile and corrective control; the language model or output alone doesn't. Taking the surrounding practice without limit would instead confer the epistemic credit of human science on every deployment. The practical limitation is familiar: information that never reaches an arrangement can't correct its output. The philosophical task is to explain why a grounding verdict leaves that limitation untouched and to distinguish inherited constraint from live correction. The decomposition earns its keep only if those distinctions improve prediction or intervention.

LLMs matter because they separate routes that human epistemic life normally bundles together. The separation is clearest in the limiting case of an arrangement whose contact with the world remains mediated mainly by previously produced human text rather than live retrieval, perceptual inputs, execution results, or action outcomes. That case supplies a baseline, not a description of every LLM arrangement. Where training preserves the effects of target-sensitive correction in source practices, it can supply DERIVATIVE ANSWERABILITY. LIVE ANSWERABILITY adds a temporal condition: a fresh discrepancy has to be able to reach current production. Relative to the baseline, retrieval, tools, code execution, multimodal input, and action-guided feedback add live routes, reducing reliance on inherited constraints for the tasks those routes can serve. Their diagnostic value depends on which route each adds and where any resulting improvement transfers. Figure 1 sets out the difference.

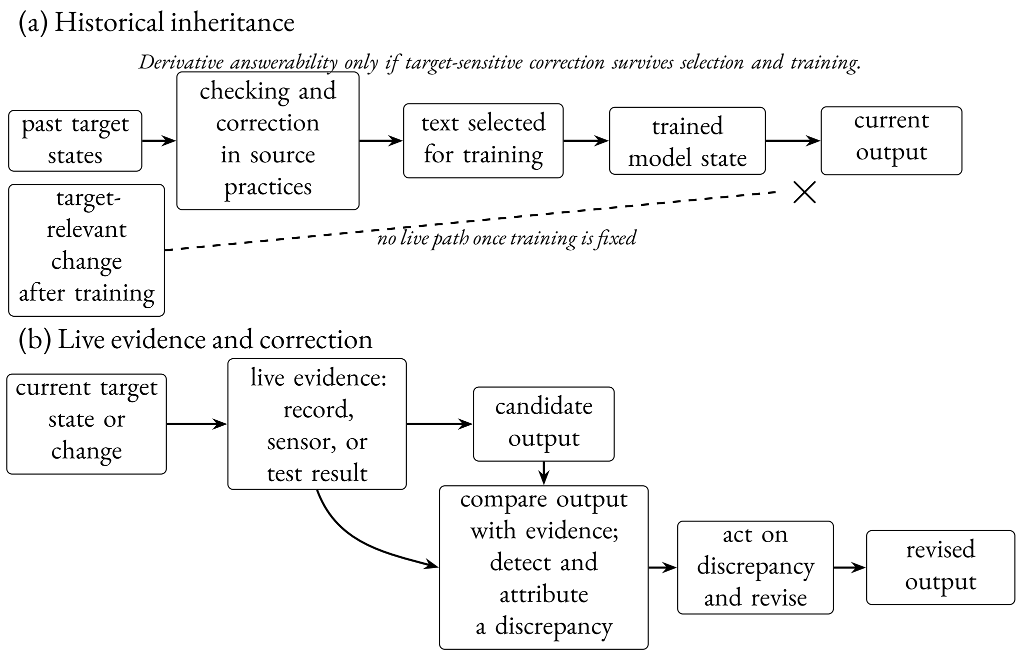

Figure 1: Historical inheritance versus live correction. Solid arrows show possible causal influence, not answerability or warrant; the dashed line ending at × shows no directed path. In panel (a), target-sensitive correction in source practices can survive selection and training, supplying derivative answerability, although a fresh target-relevant change can't reach current output once training is fixed. In panel (b), live evidence constrains a candidate, enabling detection, attribution, and revision. This directed path supplies live answerability; it supplies corrective control only when evidence reaches the arrangement, checking is sufficiently independent, the discrepancy is attributable, and revision improves the output. Section 4 separates those conditions into five graded features.

An intervention should help only where it supplies a route the task needs. Self-consistency should help most where the missing support is coherence; retrieval should help most where the missing support is record access. Feedback based on human preference judgements should improve conversational fit before it improves answerability. Multimodal input should help some perceptual tasks before it helps calibrated measurement. These contrasts provide an initial test of the proposed distinctions among routes. Indiscriminate transfer would count against the proposed decomposition.

These contrasts compare epistemic routes while leaving open what truth consists in. Correspondence, coherence, pragmatism, reliabilism, and social epistemology each identify a relation or process that can constrain representations in ways relevant to getting things right. Deflationism leaves those explanatory questions open while denying that truth itself requires a substantive reduction. The comparison concerns how these relations and processes combine within an arrangement and what follows when one is missing. It's a framework for comparison, not a reconciliation of the theories.

This paper develops that account of epistemic architecture in three steps. First, it isolates the live, task-indexed routes by which a fresh discrepancy reaches production as something to be explained in its own right rather than a corollary of content. A positive grounding verdict can settle what a system's representations mean or refer to without licensing an inference about which fresh errors a deployed arrangement can catch.

Second, it separates the profile from corrective control. Route identity and coupling determine the kind of target constraint available and where it should help. Five arrangement-level features then diagnose whether target-relevant variation can expose a sufficiently independent and attributable discrepancy and produce a target-improving revision. They also separate the capacity to correct from observed correction performance.

Third, it turns the full decomposition into predictions that can fail. Route-by-task interactions test whether the proposed routes track selective transfer. An information account can predict them too by appealing to the task-relevant material available to the arrangement. The stronger test is held-out transport to new cases. A decomposition fixed in advance should predict new combinations of routes, tasks, and perturbations better than a prespecified model of the task-relevant information supplied to an arrangement. It should also guide intervention choice without conceptual refitting.

Sections 2–4 fix the target, explain route profiles, and define the corrective-control features. Section 5 applies the decomposition to language-model arrangements and derives the empirical transfer tests. Section 6 compares it with component theories and addresses the principal objections.

2 The target: truth-relevant representational success

Before an arrangement's profile can support projection, its target has to be fixed. This paper doesn't profile truth considered as an abstract property or as the role of the truth predicate, neither of which has perceptual, testimonial, or instrumental mechanisms as parts. It profiles TRUTH-RELEVANT REPRESENTATIONAL SUCCESS: beliefs, assertions, inscriptions, measurements, models, and inferential outputs assessed as getting things right.

That target extends beyond central TRUTHBEARERS. Beliefs and assertions can be true or false in the ordinary sense. Neighbouring cases support truth-ascription by yielding, recording, or constraining claims: a measurement, map, statistical model, or experimental display may be accurate, calibrated, or well-fitted in ways that make associated propositions true.

These cases often divide the representational labour. In a lead-exposure case, the assay reading, calibration log, database entry, and clinician's assertion do different jobs. Only the last may assert that the sample contains 3.2 mg of lead, but the others make that assertion available, checkable, and defeasible. A map can likewise preserve spatial relations that support claims about location, and a model can constrain which inferences remain live. The object of analysis is the background through which such outputs support getting things right.

Fixing the target this way leaves a reductive definition of truth to one side. Deflationary and minimalist theories can give the truth predicate a thin logical or generalizing role while keeping a substantive causal story out of the basic theory of truth (Horwich, 1998). That point can stand. What remains to explain is why truth-relevant representational successes cluster and fail in patterned ways, and when evidence about them supports expectations about nearby cases.

The target leaves correspondence intact. Correspondence theories preserve the idea that truth involves answerability to how things are (David, 1994; Kirkham, 1992). Correspondence fixes the target of answerability; it doesn't specify the supporting routes.

ANSWERABILITY generalizes the constraint beyond central truthbearers. A representation is answerable to a target when discrepancies between it and the target can bear on its production, acceptance, correction, or withdrawal. Strictly, profiles belong to arrangements, and an output is answerable relative to the arrangement that produces, maintains, and revises it. For brevity, I sometimes call an output answerable without restating the arrangement that supports the relation.[3]

Answerability is graded because routes differ. A calibrated instrument is answerable when checks, recalibrations, and downstream failures can expose discrepancies. A coherent but disconnected arrangement is less answerable when the relevant fact has no route into production. Audit, replication, practical failure, or later correction can still connect a fact to an output, even after delay. Specific, repeatable routes that readily expose the relevant error make that connection stronger; generic or loosely targeted routes make it weaker.

Different routes make the gradation concrete. Repeated weighing, standard weights, and failed downstream uses can expose scale drift. Records, other witnesses, and later events can defeat a mistaken report. A language model that generates a plausible citation from textual pattern alone has so far been shaped by genre, with those checks absent from the production route. Its citation may still be right, but its rightness is less supported by the route that produced it.

Code execution supplies a contrasting live route. A language model can write a function, run it, inspect a compiler error or failed test, and revise the code. A one-shot generator lacks that route even when its first answer happens to work.

The contrast survives when content is held fixed. *The sample contains 3.2 mg of lead* is strongly answerable when it's generated by a calibrated assay whose logs, standards, and downstream checks can expose error. The same string is weakly answerable when a text-only arrangement produces it because such numbers fit the genre. The truth conditions haven't changed; the route by which the claim became available has.

Pragmatist theories identify practical correction as another route to answerability. Inquiry, action, and successful practice can correct locally coherent or socially inherited representations with weak world-connection (Dewey, 1941; James, 1907). That doesn't make truth identical with what works. Practical success helps arrangements become answerable to their targets.

Here, TRUTH-TRACKING has a modest sense. It names patterned support for representational success, whereas Nozick's (1981) truth-tracking account specifies subjunctive conditions on knowledge. Patterned support lets us expect some representational successes to survive checks, perturbations, and neighbouring applications better than others.

Boyd's (2021) accommodationism explains why such support can be patterned. Reference and truth depend on more than definitions in isolation; they're achievements of practices that align perceptual, instrumental, cognitive, and representational activity with relevant causal structures. That account raises a narrower diagnostic question: how strongly do those routes and practices constrain a given arrangement?

For LLMs, that diagnostic asks how much of this background a given arrangement inherits or instantiates. A text-only arrangement may inherit patterns shaped by testimony, statistical regularities, and local coherence constraints while lacking live perception, measurement, intervention, and ordinary practical correction. That profile makes truth-relevant assessment possible. The decomposition predicts uneven, empirically diagnostic success.

With the target now fixed, which route patterns support projection to nearby cases?

[3] The lexical history of *true* is illustrative, not evidential. The *Oxford English Dictionary* records senses of loyalty, veracity, accordance with fact, accurate fit, and mechanical alignment (Oxford English Dictionary, 2026). They give the broader constraint relation historical precedent, but inferring the analysis from them would be an etymological fallacy.

3 Routes and profiles

Goodman's (1955) new riddle asks which predicates support PROJECTION from known to unknown cases. Here, PROJECTIBILITY concerns whether evidence about an arrangement's route profile warrants expectations about its likely success or failure on nearby tasks.

Khalidi's (2013) account provides the causal-network framework for this use of the profile. Natural categories earn their status by supporting prediction and explanation, and they do so when associated properties stand in causal relations that support projections from some properties to others.

Applied here, Khalidi's framework distinguishes the evidence that warrants a projection from the world-side relations that make it reliable. Relevant routes include perception, testimony and records, coherence constraints, measurement, intervention, checking and revision, and feedback from practical success or failure. Together, these routes make representational outputs more or less answerable to the world. Khalidi's later formulation of natural kinds as "nodes in causal networks" supplies the graph vocabulary (Khalidi, 2018, p. 1387). On his account a kind is a node, a highly connected vertex whose core properties give rise to derivative ones, and the network is what the node sits in. A profile isn't a kind in that sense. I take from him the discipline that connectivity is the proposed world-side ground of the projections, not a claim that truth-tracking is a node with a core.

Routes do different jobs. Some bring target information into the arrangement, some integrate it with existing representations, some expose discrepancies, and some return practical failure to later production. A route can do more than one of these jobs. The routes' corrective significance depends on whether, together, they let the arrangement detect and repair a fresh discrepancy.

Calling the result a profile marks two points. First, different routes can dominate in different domains. Proof norms matter more in mathematics than ordinary perception does; calibration matters more in laboratory measurement than conversational fluency does. Second, the epistemic significance often lies in the relations among routes. Testimony matters differently when it's backed by records, instruments, and correction than when it's an isolated report.

Part of this decomposition comes from Boyd's property-cluster account (Boyd, 1991, 1999). Causally related property clusters can make real kinds support projection without essences. Boyd's (2021) accommodationism adds the detail that matters here. Perceptual skills, instruments, inferential routines, social arrangements, and representational uses each contribute when they help a practice align with relevant causal structures. The routes form a coupled pattern, not coordinate epistemic goods.

Accommodation, in this sense, is the gradual fitting of representational practice to the structures it's used to track. Practices of using scientific terms, producing instrument readings, and applying ordinary object names become more reliable as perception, measurement, inference, correction, and use are adjusted together. Definitions alone don't secure the fit; practices that make error detectable and revision possible maintain it.

Consider *fever*. Feeling hot, thermometer readings, calibration standards, thresholds, records, and revised estimates of normal body temperature have adjusted one another over time. The term remains useful because these routes expose error and let the practice revise its measurements.

Applied to text-trained arrangements, the analysis remains continuous with Boyd at the explanatory level and diagnostic at the arrangement level. Accommodationism tells us why representational practices can be answerable to causal structure; the profile records how strongly the relevant routes constrain a given arrangement. A text-trained arrangement can inherit stable linguistic and testimonial patterns while its instrumental, perceptual, and practical routes remain attenuated.

In arrangements for everyday perception and basic coordination, perception, action, memory, testimony, and practical failure – the routes where error has been costly longest – have repeatedly corrected one another. A language-model arrangement that lacks perceptual and verification routes should show a corresponding failure pattern.

In arrangements for frontier science and remote historical fact, the routes may instead be sparsely linked. Error cost is delayed, diffuse, or beyond current instruments. There may be inference norms, expert testimony, and coherence constraints, but fewer shared routes of correction. Shared selection or training history, accumulated correction procedures, and measurement infrastructure are often limited there too.

Network boundaries still depend on the chosen level of abstraction. Onishi and Serpico (2022) show this for causal graphs, and Craver (2009) makes the same point for mechanism talk. Following Khalidi's (2013) distinction between epistemic purposes and non-epistemic practical or aesthetic ones, the purpose here is to represent a target domain well enough for relevant inferential, practical, or corrective success under perturbation. Connections can be sparse without dissolving an arrangement's profile relative to that purpose.

4 Corrective control and bounded projections

Routes and their relations define the profile, but they don't by themselves warrant projections about further truth-relevant success or failure.

Perturbation makes the declared purpose empirically testable. It distinguishes lucky fit from robust answerability. A claim based only on success in the training cases, the familiar environment, or the current conversational context supports only narrow projections. It earns stronger warrant when the representation continues to guide action, inference, measurement, and correction after conditions change.

Testing that persistence requires only a modest causal framework. Pearl (2009, 2010) distinguishes association, intervention, and counterfactuals. Observed route identities and relations describe the initial profile. Perturbation helps identify mechanism, and projectibility asks counterfactually what would hold nearby. Woodward (2001, 2003) supplies the manipulability reading: a route is explanatory when interventions on it change performance in stable, predictable ways. The argument doesn't require a fully specified structural causal model.

What matters is invariance under the declared changes. A route pattern can support bounded expectations without being actively maintained by correction. Corrective control is one way such robustness can be produced, but neither past stability nor projectibility by itself establishes live error detection and repair(Reynolds, 2026).

Corrective control adds capacities for discrepancy detection and target-improving change. An arrangement has corrective control only relative to a target and perturbation range. Its strength within those bounds depends on five features of the arrangement's corrective architecture:

1. Target access: target-relevant variation reaches the arrangement through an operative route.
2. Detectability: a discrepancy can be registered by that arrangement against a standard it can consult.
3. Checking-route independence: the checking signal doesn't merely reproduce the production route's error.

4. Attribution: the discrepancy is specific enough to guide a response.
5. Effective uptake: the capacity to make a target-improving change early enough when a relevant discrepancy occurs, is detected, and is attributable.

Together, the five features specify when the live path in Figure 1(b) supplies corrective control rather than merely carrying information.

These are graded, potentially dependent dimensions; the numbering is for reference, not rank. Target access, for example, belongs to the arrangement rather than to a sensor, record, or checker considered alone: a route may be present without transmitting the relevant variation for the declared task. Likewise, a checker's errors may partly correlate with the generator's, attribution may be more or less precise, and uptake may be delayed or probabilistic.

A single architectural change can improve several features. Giving a checker access to repository state, for example, can improve target access and reduce its error correlation with the generator. This account ranks one arrangement above another when it instantiates every feature at least as fully and some more fully. Trade-offs needn't be rankable: greater checking-route independence can come with weaker attribution. Nothing here licenses a weighted score across the five.

These five features characterize corrective capacity, not observed rates. Three quantities differ: architectural capacity; exercise frequency, or how often a route actually alters production; and conditional repair performance, or how well the arrangement responds when a relevant discrepancy is present. Effective uptake isn't exercise frequency. Otherwise, a near-flawless arrangement in a benign environment would score low, while a poor generator triggering a mediocre checker would score high. Counting any alteration also rewards changes in the wrong direction.

Effective uptake instead asks a conditional, directional question: once a discrepancy is detected and attributable, can the arrangement reduce it? Conditional repair performance is the probability that it makes an appropriate target-improving change when a relevant discrepancy is present, considered alongside its rate of unnecessary or harmful intervention when correction isn't warranted. Detection, exercise, and uptake after detection can be reported as component rates. Audited end-to-end correction and false-intervention rates provide evidence about capacity and observed performance. Raw repair frequency isn't what corrective control consists in.

Among the five, checking-route independence alone compares one route's errors with another's. It depends on where a checker came from, what it can see, what it's optimizing, and how its errors covary with the generator's. Target access, detectability, or effective uptake can be present with it or without it.

These conditions fix the relation to answerability rather than competing with it. Answerability comes in degrees, and its strength depends on how specific, repeatable, and proximate the route is. Corrective control is stronger than answerability alone: it requires both discrepancy detection and an effective route to repair. So an arrangement can be answerable in some measure while lacking corrective control over the targets at issue.

A thermometer connected to an independently calibrated drift monitor illustrates the distinction. Suppose it consults a separate standard, raises a correct alert, and every alert is ignored. Target access, detectability, checking-route independence, and attribution are present. Effective uptake is absent: no route connects the alert to what the arrangement does. Because the people ignoring it are part of the arrangement, that's a missing capacity rather than a low correction rate.

Route structure also supports a bounded projection. A value produced through a live assay, calibration log, and error-reporting practice warrants stronger expectations about nearby cases than one copied from yesterday's database, because changes in the sample or instrument can affect the former. Which further successes are warranted remains relative to the target, task, and perturbation.

5 Language-model arrangements: inherited structure and live routes

5.1 The arrangement and its routes

Applying the framework begins by fixing the output-producing arrangement for the target and task. It comprises the language model and prompt, together with any retrieved records, tools, sensors, checkers, human reviewers, or update routes that are live.

Multimodal foundation models are included when their linguistic outputs are under assessment. Multimodal training can supply inherited perceptual structure; a runtime image, audio, video, or sensor input adds a live route only when it's available and integrated for the task.

Retrieval belongs to the arrangement when it runs, not to the language model. Human appropriation changes the arrangement rather than adding a gloss to a fixed one.[4] The effects of training history belong to the arrangement's causal history.

The application separates inherited constraints from live routes. Training can shape production without giving the arrangement present access to the source practice. A live route can transmit a new record, measurement, or check, and it may also support correction. A purported route that supplies no target-relevant constraint is decorative: the signs of experiment, source use, or measurement remain while the constraint tends to zero. The comparison is interventionist: changing the active routes should change the expected pattern of success and failure.

Live correction also has a horizon. In a stateless chat, a correct, attributable counterexample that exposes a relevant discrepancy and causes a revision supplies effective uptake within the session and nothing beyond it. The weights don't move, the next session starts where the last one began, and no other user benefits. The arrangement has corrective control for one conversation, not for the policy deployed in later conversations.[5]

Within each profile, the five features from Section 4 determine whether the live routes supply corrective control.

Training can supply derivative answerability, but only on a condition. Discrepancies already registered in the source practices have to shape present production through the training history. Causal descent isn't enough by itself. Dense propaganda can leave distributional traces as persistent as those of a measurement literature. A language model trained on it can inherit the shape of a practice whose strongest corrective pressures track audience response, party discipline, or persuasive success rather than the empirical target at issue. What has to survive the training history is the effect of corrections made against that target rather than against audience response, rater approval, or a party line.

Three things come apart here: causal ancestry, which any corpus included in training supplies; derivative answerability, which requires target-sensitive inherited correction; and live answerability, which requires a route from a fresh discrepancy to current production (Figure 1).

[4] Preparation of this paper supplies mundane examples. When a coding agent reported sources unavailable, directing it to a local reference library added a live source route. When criticism was retained in the conversation and the author required revision, human review became a live correction route. Neither belonged to the language model alone.

[5] External state can extend this horizon without changing model weights. Claude Code's project memory and OpenAI's persistent conversation state become live routes when later sessions load them and they affect production (Anthropic, 2026a; OpenAI, 2026).

Derivative answerability by itself never supplies live answerability. Once the training history is fixed, a discrepancy arising after it has no way to reach production. Training against verifiable rewards tests this boundary. When a language model is optimized against unit tests, a proof checker, or execution results, a discrepancy between output and an independently specified target is detected and does alter production. All five features can be instantiated while the verifier runs, to the extent that it's adequate to the target, sufficiently independent of the generator, and attributable in its signal. Tests can be incomplete, attribution coarse, and verifier errors correlated with the generator's.

The training loop has then exercised corrective control over model outputs; the deployed arrangement isn't thereby correctable. The loop supplies that control only over the targets its verifier covers and only while it runs. A language model that has learned arithmetic against a checker still can't tell you whether today's figure is right. Verifiable rewards can strengthen derivative answerability, since the corrections they encode were made against an independently specified target rather than against approval alone. They don't make it live.

A live route can supply target access. It supports corrective control only if the arrangement can detect and attribute a discrepancy, receive a sufficiently independent signal, and act on it. A merely decorative invocation resembles such a route while supplying none of these features. Checking-route independence depends on the relation between routes, not on one route in isolation.

Corrective control doesn't extend beyond the arrangement's live routes, and it's bounded target by target. A user who checks a claim can supply corrective capacity for that claim. Checking frequency fixes the route's coverage across claims; appropriate improvement when a relevant discrepancy is present, considered with harmful intervention, fixes conditional repair performance. The ordinary case isn't an arrangement with no corrective control anywhere, but one asked about targets that none of its live routes can independently check. That boundary doesn't affect grounding, but corrective control extends no further.

5.2 Route-specific interventions

Retrieval augmentation queries an external corpus or search system and places retrieved material into the language model's context; it adds record access when it works. Its characteristic failures are diagnostic rather than incidental: a stale index supplies a record of the wrong time, a chunk boundary supplies half a record, and an embedding match on wording rather than content supplies a passage that resembles the answer. Each leaves the archival route nominally present and materially absent: decorative retrieval implemented by a real mechanism rather than produced by a turn of phrase.

Tool use routes a subtask through a calculator, database, code interpreter, application programming interface, or instrument; it adds calculation or measurement only when the right tool is called and its result is integrated. When execution results guide revision, tool use adds a corrective loop.

Self-consistency prompting samples multiple completions or reasoning paths and privileges convergence among them; it strengthens internal convergence. An external check requires a route to the relevant facts whose signal isn't generated by the same convergence process.

Instruction tuning and reinforcement learning from human feedback (RLHF) change a language model's policy through demonstrations, preferences, benchmarks, or evaluations; they add a social or evaluative correction route whose target depends on the feedback.[6]

[6] Preference signals have factual targets. Rater approval of affirmation, deference, or confident presentation is a fact about a population and interaction setting, and RLHF can make the resulting arrangement's behaviour strongly answer-

Multimodal input supplies image, audio, video, or sensor information; by itself, it adds perceptual input. Calibrated measurement is a separate route.

Multimodality matters because it seems to answer Harnad's (1990) sensorimotor prescription. A finer-grained diagnosis separates perception from action, correction, and use. An arrangement built around a vision-language model trained on image-caption corpora inherits perceptual structure only through selected, framed, and captioned images. A deployment-time image input adds a live perceptual route; calibrated measurement, intervention, and action-guided correction remain separate routes.

Multimodal input should improve performance unevenly. It should reduce some failures in object classification and leave measurement- and intervention-dependent tasks fragile. It should also leave room for object hallucination, where generated descriptions include objects not present in the image (Y. Li et al., 2023). That pattern is what this account expects when target access is strengthened and the correction routes are left as they were.

These interventions can be combined, so the profiles summarized below are neither mutually exclusive product classes nor a scale. The text-only row remains a task-relative limiting case. Even that limiting arrangement inherits reports written by perceivers, explanations corrected by teachers, institutional records, and descriptions shaped by instruments. It can support genuine epistemic achievements while occupying a different role from a speaker or hearer in a testimonial exchange.

Table 1 summarizes the architecture-sensitive predictions.

able to that fact. The failure mode is target mismatch: the policy can track the rater distribution accurately while its assertions are assessed against a different, object-level target. Sycophancy is the clean case, where an arrangement accurately tracks the wrong target.

Arrangement type	Principal architectural resources	Directional prediction
Text-only arrangement with a pretrained language model	Training-distribution regularities and local coherence	Strong where dense, stable text is a good proxy; fragile for recent, obscure, perceptual, or measurement-dependent facts.
Instruction-tuned or RLHF arrangement	Rater feedback, demonstrations, and benchmark evaluation	Better conversational fit; truth-relevant gains should concentrate where raters or benchmarks have target-sensitive evidence. Independently informative checking routes still have to be supplied.
Retrieval-augmented arrangement	Live access to records and institutional sources	Better on updated or source-backed recall; remaining fragility where the missing route is experiment, perception, or intervention.
Tool-using or coding arrangement	Calculators, databases, code execution, instruments, or application programming interfaces	Better where the right tool is called and interpreted; strongest when runtime or test results guide revision; new failures from tool choice, stale data, and test-as-proxy effects.
Multimodal arrangement	Perceptual input in image, audio, video, or sensor form	Better on perceptually anchored tasks when runtime input bears on the target; no general gain on calibrated measurement or practical correction.
Action-guided or expert-reviewed arrangement	Practical feedback, review, and downstream failure signals	Stronger corrective capacity where feedback is timely and domain-competent; weaker where review is sparse or proxy-based.

Table 1: Architecture-sensitive predictions. Entries summarize expected directions; the testable claims are the prespecified route-by-task interactions.

5.3 Inherited structure and missing anchors

Textual inheritance explains both the competence and the fragility of text-only arrangements. Human text is saturated with world-directed practices of looking, measuring, arguing, repairing, citing, and correcting. The deployed arrangement usually remains outside those practices when it produces a fresh answer. It draws on their products with attenuated access to their present checks.

Text-only training can still produce more structure than a bare *textual residue* slogan suggests. The generative pretrained transformer model Othello-GPT is a challenge case: a sequence model trained to predict legal moves in a board game developed an internal representation of board state, and interventions on that representation affected its outputs (K. Li et al., 2023). The result shows that inherited constraint can be strong when the target state is densely and systematically encoded in the sequence.

Othello provides only a limited basis for a broader claim. The game is finite, fully observable, and rule-governed; its legal moves are a deterministic function of a state recoverable from the move sequence. That's why the result is both striking and unrepresentative. Nothing follows about domains whose relevant state is absent from the record, which is the class this argument concerns.

Internal world-model evidence should be strongest where the relevant state variables are encoded in the sequence and the loss function penalizes violations of their structure. It should weaken when success depends on variables absent from the sequence, live measurement, changing local conditions, or intervention in the target system. Successful transfer across those boundaries would count against this diagnosis.

Coherence also constrains these arrangements. Training and decoding reward genre fit, local well-formedness, smooth inference, and continuation from context, supporting the formal linguistic competence distinguished from functional world use by Mahowald et al. (2024). The same pressures can produce a plausible lab result, legal summary, observation, or citation without a live experiment, court record, observation, or catalogue entry.

That preservation condition predicts that textual inheritance should preserve distributional patterns more consistently than archival particulars. Particular strings, names, and sources can survive, so the claim isn't that token-bound information never does. Training supplies no general relation of record identity, provenance, and current checkability. A text-only language model can encode statistical regularities in testimonial practice. The resulting arrangement can handle a domain whose records are dense and stable while fabricating the verbatim anchors (quotations, identifiers, author lists) on which the practice's correction routines depend. The same distinction predicts that these failures should be especially retrieval-sensitive: retrieval adds the archival route that distributional inheritance lacks.

Hallucination plays a more specific role here. The LLM literature treats it as heterogeneous (Huang et al., 2025). The narrower claim is that text-only arrangements should be especially vulnerable to fluent, coherent, genre-appropriate outputs that fail when the task requires a fresh or independently informative route to the facts. Those failures should be common for obscure sources, recent events, local observations, novel measurements, fine-grained quantities, and action-sensitive causal claims. They should be rarer where dense, stable, redundant text is a good proxy.

Frankfurtian accounts of bullshit (Frankfurt, 1986) diagnose LLM output differently. Hicks et al. (2024) apply that diagnosis to ChatGPT-style systems, arguing that their outputs are produced with indifference to truth. On the Frankfurtian diagnosis, the system doesn't care where the facts are; on this diagnosis, the relevant facts have no route into production or correction. That missing route yields comparative predictions across retrieval, tools, multimodality, and action-guided feedback. This diagnosis asks for a route, not a better attitude.

Mitigation results should be attributed to the route manipulated. Evidence that retrieval reduces hallucination in conversation bears on record access, not on perception, experiment, or intervention (Shuster et al., 2021).

Action-guided arrangements can substantially expand corrective control when the feedback is timely, attributable, and tied to the relevant failure. The code-execution case in Section 2 supplies the software example. A robot that breaks a glass, misses a target, or fails a laboratory protocol can receive similarly direct correction; a chatbot receiving delayed approval, engagement metrics, or sparse user edits has a slower, less attributable route. The label *feedback* hides that difference unless the route is specified.

5.4 What the correction routes correct

Instruction tuning, RLHF, benchmark evaluation, red-teaming, and expert review function as correction routes, and they should be classified by what they correct. Their immediate targets often include helpfulness, harmlessness, preference satisfaction, conversational fit, and benchmark success. Those constraints can reduce some errors and can also reward answer-shaped behaviour when users or raters prefer confidence, agreement, or plausible explanation.

The verifier case also separates two forms of post-training. Rewards based on a rater's approval target that approval; verifier-based rewards target an independently specified criterion, subject to the verifier's adequacy. The label *post-training* covers both, but they build different profiles. Treating checker-based reward as mere rater approval understates what it supplies; treating rater approval as truth-directed feedback overstates it. For this analysis, the targets against which rewards were computed matter more than the amount of post-training.

Work on sycophancy shows the risk of treating feedback based on human preference judgements as truth-directed on its own (Shapira et al., 2026). These processes strengthen some social and evaluative constraints while leaving routes through perception, measurement, and intervention indirect or absent.

Accounts of fine-tuning disagree because they seek to explain different things. Grindrod (2024) treats it as a relatively marginal procedure that can't be what makes the difference between intentionality and its absence, since language models produce apparently meaningful text without it. Ruyant (2026) treats it as constitutive of what the system represents, because the norms of appropriateness a language model is tuned toward are what fix its target.

Fine-tuning can be constitutive of appropriateness for a purpose, the phenomenon Ruyant seeks to explain, because norm setters fix that purpose. It can remain marginal to whether outputs are meaningful in Grindrod's sense. Neither verdict settles target-directed correction. Rater approval supplies that only when the judgement is informed by a target-sensitive route: fact-checking against records, tool results, or action outcomes rather than approval alone. Treating all three questions as one quantity called *grounding* makes the disagreement look unresolvable.

5.5 What grounding leaves open

Referential and teleosemantic arguments can establish genuine, though limited, content in LLM states (Coelho Mollo & Millière, 2026; Mallory, 2026). That conclusion isn't in dispute. The remaining issue is whether the deployed arrangement has the live correction, measurement, intervention, perception, or feedback routes required by the task.

Mandelkern and Linzen (2024) illustrate the dissociation. They close with Izzy, isolated from birth in a sensory-deprivation chamber and reaching the world only through a text screen, and they argue that externalism dissolves the puzzle about how his words refer. Izzy's answerability is nonetheless confined to the testimonial and inferential routes his screen supplies, with perception, measurement, and intervention absent. Secure reference alongside a restricted profile is the dissociation at issue.

Pepp raises a stronger version (Pepp, 2025). She suggests that such systems may contact objects through their exchanges with users, coming into contact with whatever shaped a user's input. The inherited–live distinction splits that case. A live image-input route supplies perceptual access. Contact with the objects that shaped a user's text is inherited: the perceiving was done by someone else, and nothing in the exchange reports back when the user was wrong.

Bender and Koller's octopus and Mandelkern and Linzen's Izzy support opposing verdicts on reference. Their shared limit is architectural. In each case, linguistic traffic is the principal route, so a discrepancy that never enters that traffic can't alter present production. What isn't on the wire can't correct what is.

5.6 Corrective control in deployment

A deployment case isolates what adding a checker changes. Anthropic (2026b) describes a coding agent that routes each proposed action through a classifier that blocks operations it judges irreversible, destructive, or aimed outside the working environment. The case concerns safety control rather than factual truth directly. It instantiates the live correction path in Figure 1: a separate discrepancy signal becomes behaviourally effective.

When its policy judgement is correct, the separate classifier can supply detection and effective gating where the acting language model registers nothing itself. Adding it can give the arrangement detectability and effective uptake through a second route rather than through introspection.

Separation alone doesn't secure checking-route independence. An external checker can reproduce the generator's errors, while the generator given independently sourced environmental information may not. Relevant dimensions include development provenance, the environmental information available to the checker, objectives, and conditional error structure. The reported hardening gave the classifier more environmental information, including repository visibility and git state. Whether that change decorrelated its errors wasn't measured, and lineage alone wouldn't settle the question. Measuring it requires conditional error rates for checker and generator on a shared held-out set.

The deployment figures illustrate the distinction between exercise frequency and conditional repair performance. Users reject 3% of individual permission requests against 39% of plans put to them for approval. Both are observed intervention frequencies, not conditional repair performance; without audited rates of unsafe proposals and false blocks, that performance is unidentified. The gap is consistent with route-specific non-exercise in the per-action arrangement, though it doesn't isolate that explanation from interface burden, action granularity, or perceived stakes.[7]

5.7 Empirical transfer

Route-specific transfer provides a component test. Improvements in one route should remain limited where a task depends on a different route. If perception, measurement, retrieval, tool use, expert correction, or action feedback is confirmed to be active and reaches production but fails to alter the expected contrast, the proposed decomposition has misdescribed the mechanisms. If text-only arrangements remain mainly dependent on inherited textual accommodation and local coherence, the same vulnerability should persist even as fluency improves.

A clinical test makes this concrete.[8] One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area and answer format and requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal

[7] A controlled study points the same way, with human testers catching 13.6% of planted dangerous commands against the classifier's 89%, though its participants worked in a purpose-built environment and knew they were being evaluated (Anthropic, 2026b). Both figures are unaudited vendor self-reports and illustrate the exercise gap rather than measuring conditional repair performance.

[8] The design parallels validity argumentation in measurement: inferences from performance are licensed relative to a purpose and tested by varying method while holding the target construct fixed (Kane, 2013; Messick, 1989).

judgement. The route decomposition predicts a retrieval gain on the record-recoverable set and little transfer to the patient-specific set unless the needed measurement or tool route is supplied. Retrieval alone producing the same gain on both sets would count against the proposed distinction between routes.

The implementation is crossed rather than merely matched: closely related item families vary whether a route is available and whether the task requires it. Substantive correctness, scored blind across both item types, is the primary outcome. Online Resource 1 specifies the estimand, controls, secondary outcome, worked decision threshold, and interpretation of uncertainty. The main test is the route-by-task interaction declared before outcomes are observed.

One interaction is weaker than two. Retrieval should help the record-dependent items, and a measurement tool should help the measurement-dependent ones, so crossing both interventions against both task types tests a double dissociation. That is, each route should confer a selective advantage on the task that needs it. The contrast tests whether the proposed routes track selective transfer, but the interaction isn't unique to the decomposition. A relevance-sensitive information account can predict the same crossing by treating records and measurements as different task-specific information.

The natural rival explains the crossing through differences in task-relevant information availability. Evaluating that reply requires distinguishing three readings of coverage. Input coverage concerns task-relevant content accessible at a declared architectural boundary before the focal component produces its signal. Depending on that boundary, the inputs can include a prompt, retrieved passage, database response, sensor output, or candidate answer presented to a checker. The boundary is declared and relevance coded against a gold-standard task analysis before system outputs are observed. Input coverage is a restricted empirical comparator: two arrangements can receive the same task information while differing in what they do with it.

Conditional coverage concerns what one route adds given what the other routes have supplied. It can represent the greater value of a checker whose errors don't reproduce the generator's. Effective coverage includes whatever information can influence the final output given the architecture, provenance, checking relations, and uptake policy. It summarizes the usable information supplied by the whole arrangement without decomposing how it became usable.

The clearest competing account uses input coverage. Conditional coverage remains an empirical comparator when its measure is fixed in advance, but it already incorporates checking-route independence and joint error structure. Effective coverage includes nearly the whole profile. Used without fixed application conditions, it can fit any result post hoc by counting each helpful route as an increase in coverage. It then summarizes the arrangement's usable information rather than competing as a mechanism. Effective coverage may serve coarse retrospective prediction while the decomposition serves intervention, audit, and transfer. Nothing here requires one uniquely correct description for both purposes.

This yields an identification limit. For any mapping from the decomposition's variables and environments to performance, an unrestricted coverage variable can be defined to reproduce that mapping. Behavioural results alone can't identify which architecture produced the usable information. Lineage, information flow, conditional error structure, and update routes supply the additional evidence. The empirical comparison needs fixed application conditions specified before the outcomes are observed.

The empirical burden falls on the proposed decomposition. Route identity and coupling specify what kind of target constraint is available. The five arrangement-level corrective features diagnose whether target-relevant variation can reach the arrangement, yield a detectable and independently informative discrepancy, be attributed, and alter behaviour in a target-improving direction. Both components are fixed from architecture and history before performance is measured.

For the projectibility claim, compare a prespecified input-coverage model with a multilevel model containing route identities, task requirements, cross-route relations, the five features, and declared interactions. Evaluate both on held-out combinations without reclassifying routes or redefining features after observing the outcomes. No weighted profile score is required, and the decomposition loses if input coverage predicts those cases equally well.

For the practical claim, the decomposition has to improve diagnosis or the choice among retrieval, measurement, independent checking, and enforced uptake beyond whatever predictive gain it supplies. Evidence about how the routes have checked or corrected one another (their calibration history) earns a place only where it helps characterize the route relations, the five features, or their stability under perturbation.

A second component test varies checking-route independence (feature 3). Compare a checker from the generator's lineage with one trained and calibrated separately while approximately matching externally supplied task information, standalone accuracy, interface, and revision policy. The decomposition predicts greater corrective value from the checker with lower conditional error overlap. That comparison operationalizes the earlier restriction on calibration history: it matters where it predicts current error structure or stability under perturbation. An input-coverage account predicts no advantage under the declared boundary; a conditional-coverage account can predict one only by representing the joint error structure that checking-route independence names. Online Resource 1 supplies the full controls and measurement boundary.

Online Resource 1 also compares a frozen record with a live route to the same source under update and no-update arms. The test operationalizes live answerability but doesn't distinguish the decomposition from dynamic coverage, which predicts the same follow-up pattern.

Failure of a declared contrast, after confirming that the relevant route was invoked and reached production in time, would count against the corresponding route distinction or corrective feature. Repeated failure across prespecified component tests would defeat the full decomposition. The deployment case changed environmental access without measuring error correlation, so it bears on one feature rather than testing the full decomposition.

Even perfect co-occurrence within a sample wouldn't show that grounding entails corrective control: both may depend on a third feature, or every sampled arrangement may satisfy more than grounding requires.

6 What the decomposition adds and principal objections

The language-model cases show what the decomposition adds to the component theories canvassed earlier. Reliabilism foregrounds reliable processes (Goldman, 1979); coherentism, integration constrained by input (BonJour, 1985); and pragmatism, correspondence, and social epistemology supply other routes and target relations already introduced. The comparison concerns how those routes interact within one arrangement and what should fail when an interaction is removed.

6.1 Reliabilism and relations among routes

Relations among routes can change the evidential value of each. Perception checked by testimony, testimony disciplined by records, and measurement corrected by intervention have a calibration history. Evidence from that history matters only where it predicts present error relations or stability under perturbation. The WHO Surgical Safety Checklist gives the larger-scale pattern: copying the checklist without the team coordination, local adaptation, coaching, and feedback involved in effective implementation need not reproduce its effects (Armstrong et al., 2022; World Health Organization, 2009).

Reliability and answerability remain distinct. Reliability is statistical; answerability is a counterfactual relation between a representation and its target, relative to an arrangement; corrective control is an architectural capacity. An ordinary accuracy record estimates the first, whereas audited correction and false-intervention rates bear on the third.

The clinical contrast in Section 5.7 also shows why process individuation matters. Describing a retrieval-backed arrangement as *a system that answers medical questions reliably* invites transfer from record-backed questions to tasks that require live measurement, patient-specific observation, calculation, or intervention. A careful reliabilist can instead individuate the process as retrieval-backed answering under a specified profile of records, tools, and checks. That yields the same verdict. The remaining difference is methodological: the decomposition has to be fixed before the transfer is granted.

At arrangement scale, this raises reliabilism's generality problem (Conee & Feldman, 1998). Pretraining, filtering, RLHF, retrieval, tools, prompts, and institutional uptake permit many process descriptions. If selection history, correction procedures, measurement infrastructure, and current architecture pick the description that supports projection, the account supplies the individuation theory. The account loses that role if those variables don't improve held-out prediction or intervention choice.

The projective basis is the arrangement's current route structure together with whatever calibration history bears on its stability under perturbation. It isn't a single process assessed apart from that background.

Route relations also matter when sources conflict. An inspector may judge by sight that a platform is level while a calibrated inclinometer puts its slope outside tolerance and technicians report that the instrument is functioning normally. No source has standing priority; the comparison depends on assessed reliability and a history of mutual correction. Agreement among sufficiently independent witnesses confirms more than agreement inherited from a common source or error process (Bovens & Hartmann, 2003; Olsson, 2005). Coherence faces the parallel limit: a false proposition can cohere with a specified set (Russell, 1907), and corpus-grounded meaning can remain detached from a live route by which target drift alters production (Havlík, 2024).

6.2 Interest-relativity

One objection questions the profile's objectivity. Causal graphs and mechanism boundaries depend on explanatory interests (Craver, 2009; Onishi & Serpico, 2022); no network fixes a unique, interest-free carving. Khalidi (2013) likewise allows overlapping scientific domains to support different categories and methods.

That concession fixes the conditions of application rather than the result. The target, task, perturbation, and failure condition are declared first; evidence then determines whether the projection succeeds. Interests select the projective claim, not its truth. Connectivity is characterized before present performance through selection or training history, accumulated correction procedures, and measurement infrastructure. Those marks can establish occasions for one route to expose another's errors; longevity or consensus alone can't.

Choosing the purpose after seeing the verdicts would make the account unfalsifiable. Prospective characterization blocks that move. A failure on the declared held-out cases requires the profile to be revised, narrowed prospectively, or abandoned rather than rescued by redescribing the routes after the fact.

6.3 Derivative testimony and fictional output

Testimony ordinarily involves assertion, responsibility, trust, uptake, and defeaters (Coady, 1992; Goldberg, 2010; Lackey, 2008). Conversational systems strain such anthropocentric roles without simply becoming ordinary testifiers (Freiman, 2024). The claim here is derivative: a text-trained language model inherits artifacts of testimonial practice, not testimonial standing. It remains outside the current exchange and supplies neither a normal hearer who trusts under standing norms nor a speaker who assumes responsibility.

That history carries epistemic support only when target-sensitive correction survives selection and training. Section 5.3 gives the resulting prediction: distributional features of a practice should be preserved more consistently than source identity, provenance, defeaters, or verbatim anchors. Causal descent without that preservation supplies influence, not derivative answerability.

A related objection treats generated outputs as fiction. Ruyant (2026) argues that a language model represents a constructed class of appropriate linguistic production and that chatbot outputs are props in a game of make-believe. His own scope condition is architectural: the generalization holds to the extent that direct worldly inputs don't train the system. He also grants that generated text becomes dissertations, emails, programs, news, or scientific prose when people appropriate it.

Appropriation needn't change the string's kind. The same string can be correct in the game, false about the world, and effective as an artifact. What changes is the target and the arrangement: measurement, review, and correction may become live when a person or institution takes up the output. Target-indexed answerability extends the fictionalist analysis rather than requiring its rejection.

6.4 Scope

Mathematics and logic lie outside this paper's world-directed target, although proof and formal checking show that detection and correction needn't be perceptual. Deflationists may deny that truth has a substantive nature (Horwich, 1998), while pluralists distinguish ways of being true across domains. Neither view conflicts with the narrower claim that evidence about different routes may warrant bounded projections among truth-relevant representational successes. The account explains those projections, not truth itself.

7 Conclusion

The octopus and Izzy support opposing verdicts on reference, but the central question survives either one. Representations can have content, reference, and correctness conditions even when the arrangement lacks live, sufficiently independent routes from detected discrepancies to altered production. Grounding doesn't entail corrective control.

Language-model arrangements expose the dissociation at scale. Their language models inherit patterns shaped by testimony and coherence constraints. Those patterns supply derivative answerability where source practices have already made target-sensitive corrections, while the arrangements may lack live routes that detect a fresh error and carry it back into production. Identifying which routes constrain an arrangement, how they're related, and for which targets shows whether the missing corrective routes have been supplied or only partly compensated for. Rich internal structure can arise where the target domain is densely encoded in the sequence stream without supplying those live routes elsewhere.

A thin truth predicate can coexist with thick truth-tracking practices. Recurrence of a profile can support bounded expectations without live correction; corrective control adds capacities for detecting and repairing fresh discrepancies. None of that architecture need be built into truth itself.

The older epistemologies keep their subject matter. Each can represent relations among the processes it describes, and reliabilists and social epistemologists routinely do. The relevant routes normally operate jointly; language models make their separation visible.

The resulting framework compares arrangements rather than assigning systems to fixed classes. Effective coverage can summarize whether an arrangement had usable information, but it doesn't by itself say which intervention to choose or where a gain should transfer.

Retrieval, multimodal input, tool use, expert review, experimental embedding, and action-guided feedback alter an arrangement's route profile. The empirical burden is to specify in advance which route each intervention adds and where improvement should transfer. The route-by-task interactions aren't unique to the decomposition. Its stronger burden is to predict held-out combinations of interventions and tasks from a fixed decomposition better than a prespecified input-coverage model. An arrangement can improve along one dimension and remain fragile along another. Routes and coupling are characterized from causal and architectural facts before performance is checked, and the predicted contrasts can fail.

Beyond LLMs, the capacity to detect and repair fresh discrepancies depends on a distributed corrective architecture. Grounding may establish what an output is about; that architecture determines whether a fresh discrepancy can prompt revision.

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Supplementary information

Online Resource 1: Proposed empirical designs. Detailed specifications for the crossed clinical design, checker-independence comparison, and temporal test of live answerability.

References

Anthropic. (2026a). *How Claude remembers your project*. Retrieved August 11, 2026, from https://code.claude.com/docs/en/memory

Anthropic. (2026b, August 7). *Auto mode is now the default in Claude Code for Pro, Max, and Team plans*. Retrieved August 8, 2026, from https://claude.com/blog/auto-mode-default-in-claude-code

Armstrong, B. A., Dutescu, I. A., Nemoy, L., Bhavsar, E., Carter, D. N., Ng, K.-D., Boet, S., Trbovich, P., & Palter, V. (2022). Effect of the surgical safety checklist on provider and patient outcomes: A systematic review. *BMJ Quality & Safety*, *31*(6), 463–478. https://doi.org/10.1136/bmjqs-2021-014361

Attah, N. O. (2025). Do language models lack communicative intentions? *Synthese*, *205*(5), 187. https://doi.org/10.1007/s11229-025-05022-6

Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. In D. Jurafsky, J. Chai, N. Schluter & J. Tetreault (Eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 5185–5198). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.463

BonJour, L. (1985). *The structure of empirical knowledge*. Harvard University Press.

Bovens, L., & Hartmann, S. (2003). *Bayesian epistemology*. Oxford University Press.

Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, *61*(1–2), 127–148. https://doi.org/10.1007/BF00385837

Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–185). MIT Press. https://doi.org/10.7551/mitpress/6396.003.0012

Boyd, R. (2021). Rethinking natural kinds, reference and truth: Towards more correspondence with reality, not less. *Synthese*, *198*(S12), S2863–S2903. https://doi.org/10.1007/s11229-019-02138-4

Chalmers, D. J. (2023). Does thought require sensory grounding? from pure thinkers to large language models. *Proceedings and Addresses of the American Philosophical Association*, *97*, 22–45. https://philpapers.org/rec/CHADTR

Coady, C. A. J. (1992). *Testimony: A philosophical study*. Oxford University Press.

Coelho Mollo, D., & Millière, R. (2026). The vector grounding problem. *Philosophy and the Mind Sciences*, *7*(1). https://doi.org/10.33735/phimisci.2026.12307

Conee, E., & Feldman, R. (1998). The generality problem for reliabilism. *Philosophical Studies*, *89*(1), 1–29. https://doi.org/10.1023/A:1004243308503

Craver, C. F. (2009). Mechanisms and natural kinds. *Philosophical Psychology*, *22*(5), 575–594. https://doi.org/10.1080/09515080903238930

David, M. (1994). *Correspondence and disquotation: An essay on the nature of truth*. Oxford University Press. https://doi.org/10.1093/oso/9780195079241.001.0001

Dewey, J. (1941). Propositions, warranted assertibility, and truth. *The Journal of Philosophy*, *38*(7), 169–186. https://doi.org/10.2307/2017978

Frankfurt, H. G. (1986). On bullshit. *Raritan*, *6*(2), 81–100.

Freiman, O. (2024). AI-testimony, conversational AIs and our anthropocentric theory of testimony. *Social Epistemology*, *38*(4), 476–490. https://doi.org/10.1080/02691728.2024.2316622

Goldberg, S. C. (2010). *Relying on others: An essay in epistemology*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199593248.001.0001

Goldman, A. I. (1979). What is justified belief? In G. S. Pappas (Ed.), *Justification and knowledge* (pp. 1–23). Springer Netherlands. https://doi.org/10.1007/978-94-009-9493-5_1

Goodman, N. (1955). *Fact, fiction, and forecast*. Harvard University Press.

Grindrod, J. (2024). Large language models and linguistic intentionality. *Synthese*, *204*(2), Article 71. https://doi.org/10.1007/s11229-024-04723-8

Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear Phenomena*, *42*(1–3), 335–346. https://doi.org/10.1016/0167-2789(90)90087-6

Havlík, V. (2024). Meaning and understanding in large language models. *Synthese*, *205*(1), 9. https://doi.org/10.1007/s11229-024-04878-4

Hicks, M. T., Humphries, J., & Slater, J. (2024). ChatGPT is bullshit. *Ethics and Information Technology*, *26*, Article 38. https://doi.org/10.1007/s10676-024-09775-5

Horwich, P. (1998). *Truth* (2nd ed.). Oxford University Press. https://doi.org/10.1093/0198752237.001.0001

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., & Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, *43*(2), Article 42, 1–55. https://doi.org/10.1145/3703155

James, W. (1907). *Pragmatism: A new name for some old ways of thinking*. Longmans, Green, Co. https://www.gutenberg.org/ebooks/5116

Kane, M. T. (2013). Validating the interpretations and uses of test scores. *Journal of Educational Measurement*, *50*(1), 1–73. https://doi.org/10.1111/jedm.12000

Khalidi, M. A. (2013). *Natural categories and human kinds: Classification in the natural and social sciences*. Cambridge University Press. https://doi.org/10.1017/CBO9780511998553

Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, *195*(4), 1379–1396. https://doi.org/10.1007/s11229-015-0841-y

Kirkham, R. L. (1992). *Theories of truth: A critical introduction*. MIT Press.

Lackey, J. (2008). *Learning from words: Testimony as a source of knowledge*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199219162.001.0001

Li, K., Hopkins, A. K., Bau, D., Viégas, F. B., Pfister, H., & Wattenberg, M. (2023). Emergent world representations: Exploring a sequence model trained on a synthetic task. *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=DeG07_TcZvT

Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, X., & Wen, J.-R. (2023). Evaluating object hallucination in large vision-language models. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 292–305. https://doi.org/10.18653/v1/2023.emnlp-main.20

Mahowald, K., Ivanova, A. A., Blank, I. A., Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2024). Dissociating language and thought in large language models. *Trends in Cognitive Sciences*, *28*(6), 517–540. https://doi.org/10.1016/j.tics.2024.01.011

Mallory, F. (2026). Teleosemantics for neural word embeddings [Advance online publication]. *Mind & Language*. https://doi.org/10.1111/mila.70037

Mandelkern, M., & Linzen, T. (2024). Do language models' words refer? *Computational Linguistics*, *50*(3), 1191–1200. https://doi.org/10.1162/coli_a_00522

Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13–103). American Council on Education; Macmillan Publishing Company.

Nozick, R. (1981). *Philosophical explanations*. Harvard University Press.

Olsson, E. J. (2005). *Against coherence: Truth, probability, and justification*. Oxford University Press. https://doi.org/10.1093/0199279993.001.0001

Onishi, Y., & Serpico, D. (2022). Homeostatic property cluster theory without homeostatic mechanisms: Two recent attempts and their costs. *Journal for General Philosophy of Science*, *53*(1), 61–82. https://doi.org/10.1007/s10838-020-09527-1

OpenAI. (2026). *Conversation state*. Retrieved August 11, 2026, from https://developers.openai.com/api/docs/guides/conversation-state

Oxford English Dictionary. (2026). *True, adj., n., adv., and int.* Oxford University Press. https://doi.org/10.1093/OED/5727260633

Pavlick, E. (2023). Symbols and grounding in large language models. *Philosophical Transactions of the Royal Society A*, *381*(2251), 20220041. https://doi.org/10.1098/rsta.2022.0041

Pearl, J. (2009). *Causality: Models, reasoning, and inference* (2nd ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511803161

Pearl, J. (2010). The foundations of causal inference. *Sociological Methodology*, *40*(1), 75–149. https://doi.org/10.1111/j.1467-9531.2010.01228.x

Pepp, J. (2025). Reference without intentions in large language models. *Inquiry*, 1–19. https://doi.org/10.1080/0020174X.2024.2448482

Reynolds, B. (2026). *Projectibility: A history and a diagnostic framework* [Manuscript archived at PhilSci-Archive, 22 July 2026, item 30582]. https://philsci-archive.pitt.edu/id/eprint/30582

Russell, B. (1907). On the nature of truth. *Proceedings of the Aristotelian Society*, *7*(1), 28–49. https://doi.org/10.1093/aristotelian/7.1.28

Ruyant, Q. (2026). What do large language models represent? *Synthese*, *207*(1), 17. https://doi.org/10.1007/s11229-025-05416-6

Shapira, I., Benade, G., & Procaccia, A. D. (2026). How RLHF amplifies sycophancy. *arXiv preprint arXiv:2602.01002*. https://doi.org/10.48550/arXiv.2602.01002

Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021). Retrieval augmentation reduces hallucination in conversation. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 3784–3803. https://doi.org/10.18653/v1/2021.findings-emnlp.320

Woodward, J. (2001). Law and explanation in Biology: Invariance is the kind of stability that matters. *Philosophy of Science*, *68*(1), 1–20. https://doi.org/10.1086/392863

Woodward, J. (2003). *Making things happen: A theory of causal explanation*. Oxford University Press. https://doi.org/10.1093/0195155270.001.0001

World Health Organization. (2009). *Implementation manual WHO surgical safety checklist 2009: Safe surgery saves lives*. World Health Organization. Geneva.